

Mid Term Report
On
METaverse: Virtual 3D Rooms and Space

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requirements for the award of the
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Declaration

This is to certify that the project report entitled “METAVERSE: Virtual 3D Room and Space” is done by me is an authentic work carried out for the partial fulfilment of the requirements for the award of the degree of Master of Computer Applications (Software Engineering), under the guidance of Assistant Professor **Dr. Ankita Sharma**.

The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

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CERTIFICATE

This is to certify that **Mr. Mohit Chaudhary (105252002522)**, a student of the Master of Computer Applications (Software Engineering) program at University School of Information, Communication Technology (USIC&T), GGSIPU is permitted to undertake the project work as outlined in this semester progress report under my supervision.

To the best of my knowledge, the work presented in this proposal is original and has not been previously submitted for the fulfilment of any degree and other academic qualification.

Signature of Supervisor

Dr. Ankita Sharma

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ACKNOWLEDGMENT

With pleasure, I take the opportunity to express my sincere thanks and obligation to my esteemed guide **Dr. Ankita Sharma**. It is because of her able and mature guidance and co-operation without which it would not have been possible for me to complete my project.

It is my pleasant duty to thank all the staff members of the University School of Information, Communication Technology (USIC&T), GGSIPU who never hesitated and guide me time to time during the project.

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Date: __/__/2025

Signature

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1. Abstract

The project Real-Meet is a 3D metaverse-based communication platform designed to create an immersive virtual environment where users can interact and collaborate in real time. The goal of the system is to simulate real-world meetings and social interactions in a shared digital space, enabling users to communicate through avatars and experience a sense of presence regardless of physical distance. The project integrates modern web technologies including React.js, Three.js, Node.js, Express.js, Socket.io, and PostgreSQL to achieve seamless synchronization between users. The 3D environment allows avatars to move, interact, and respond to real-time socket events. A secure JWT-based authentication system ensures user identity management, while modular backend and database structures support scalability. At this stage, the core architecture of Real-Meet has been successfully implemented — including environment setup, real-time socket connectivity, authentication, room management, and synchronized avatar movement. Ongoing development focuses on enhancing the realism of the 3D world, designing more detailed rooms and avatars, and integrating WebRTC for audio-video communication.

The progress achieved so far establishes a strong technical foundation for a fully interactive metaverse platform that bridges virtual collaboration and real-time communication in a browser-based ecosystem.

2. Introduction

The rapid shift towards digital communication platform has demanded a need for an interactive way of handling meetings which could feel like real unlike other classical conferencing apps who lacks sense of presence of found in real physical environment.

This Project is all about building a 3D metaverse where online personal meetings or online classrooms could be held in a more interactive environment which could feel like a real world. This also provides user with a virtual playground and free roam option, where their virtual world avatars could freely roam in the space and interact with each other in real time like some real persons do through the communication modes of text chat, voice and video if they both agree.

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2.1. Problem Statement

Existing Online Meeting Platforms provide limited engagement among the users like one could only view one's video in a screen that does not feel like they really exists which makes the virtual meetings or a learning class looks dull, boring and less effective. There is need for metaverse based system that provides real-world like interaction :-

1. Free Roam in virtual space.
2. Avatars as humans.
3. Interactive meetings and classrooms.

2.2. Objective

The main Objective of making this project is :-

1. To design and develop a 3D metaverse platform with free space to roam.
2. To enable avatar creating and customization for user.
3. To Provide meeting rooms, classrooms and virtual playground.
4. To Support real time chatting though voice, text and video chat.
5. To allow user (admin) to host a meeting, class, seminars etc.
6. To ensure Security and cross platform accessibility through web browsers with enhanced security.
7. To create a clean and responsive UI for users ease of use and manage through their system.

2.3. Scope of Project

1. The System will be served for students, workers, teachers, organizers etc.
2. Users will be able to:-
 - Create Avatars and attend meetings, classroom and events.
 - Can Also become a host of any meeting, event or classroom.
 - Communicate Through Text, voice and video.
 - Able to roam freely in a playground and interact with strangers.
3. Host will be able:-
 - First of they will be able to create their own meeting, classroom or other virtual spaces.
 - Can Allow or reject entry to any user.
 - Can also restrict motion and interactions of the avatars when the host wants.

3. Literature Review

A metaverse is virtual world in which users interact while represented by avatars typically in a 3D display, with the experience focused on social and economic connection.

The term *metaverse* originated in the 1992 science fiction novel *Snow Crash* as a blend of "meta" and "universe". In *Snow Crash*, the metaverse is envisioned as a version of the internet that is a single, universal, and immersive virtual world, facilitated by the use of virtual reality. While, A uniform Scientific definitions is not yet been found. Definitions differ from date by date and from person to person depending on their perspective.

We can't predict what internet will look aster some years but metaverse can be an answer to it. The metaverse give us a way to visualize the internet in form of a 3D space. Many of the companies have invested huge on project related metaverse mainly after the boom of the WEB3. Second Life (Linden Lab, 2003) introduced virtual economies, avatar-based interaction, and user-generated content. It demonstrated the feasibility of large-scale online social spaces. VRChat and Rec Room allow users to interact using customized avatars in real-time 3D environments, supporting text, voice, and gestures. one of the largest corporations worldwide, Facebook announced in 2021 to start the next digital revolution in *metaverse* and subsequently renamed their name to metaverse. With this event, the term metaverse gained a renewed and more strong momentum that went beyond the subject of scientific studies more towards bringing it to life. This revolution in metaverse in going to bring new areas of business and more and learnings.

Reading the research paper of [Georg David Ritterbush](#) on "Defining The metaverse" I got to their findings. They found "*Metaverse, a crossword of **meta** (meaning transcendency) and universe, describes a three-dimensional online environment in which users represented by avatars interact with each other in virtual spaces decoupled from the real physical world.* This more open definition allows a comprehensive application of the term Metaverse without omitting the most crucial foundational characteristics found in the world analysis.

The development of a metaverse is not easy and much challenging. Metaverse need performance Optimization, Rendering Complex scenes is computationally expensive. Ensuring Smooth experience across desktop, mobile etc remains challenging.

4. Theoretical Background

The concept of the Metaverse represents a convergence of physical and digital realities, enabling users to interact, communicate, and collaborate in shared virtual spaces. It integrates various modern technologies such as web-based 3D/2D graphics, real-time communication systems, and social interaction models to simulate real-world experiences within a digital environment.

The idea of creating virtual meeting spaces stems from the evolution of the Internet from static web pages (Web 1.0) to dynamic, interactive applications (Web 2.0), and now toward immersive, persistent digital environments (Web 3.0 and beyond). These environments emphasize presence, continuity, and interaction—core aspects that make users feel connected despite being geographically apart.

Real-Meet is designed as a 2D Metaverse platform that allows users to create and join virtual spaces for communication and collaboration. Each user is represented by an avatar that can navigate across virtual rooms, interact with other avatars, and initiate real-time communication such as text chat, voice messaging, and video calls.

The project makes use of real-time web technologies, particularly Socket.io, to ensure seamless communication between users without delays. The front-end is built using React.js, providing a responsive and interactive user interface, while the Node.js/Express back-end handles socket events, authentication, and data synchronization. A PostgreSQL database stores user profiles, room information, and communication logs.

By combining these technologies, Real-Meet aims to simulate the feeling of “being together” in a shared digital space. It demonstrates how metaverse-based communication platforms can enhance virtual collaboration, reduce physical boundaries, and provide new opportunities for online interaction and teamwork.

5. Work Done

The Real-Meet (3D Metaverse Communication Platform) project has progressed significantly since its initiation. The foundational components for both the frontend and backend have been developed, ensuring smooth communication between users in a shared 3D virtual space. The completed work so far includes environment setup, socket integration, authentication, and avatar synchronization.

Environment Setup

- Configured the React.js environment with Three.js (React Three Fiber) for rendering 3D scenes.
- Installed and set up Tailwind CSS for responsive and modern UI design.
- Initialized Node.js and Express.js backend with API routing and socket server configuration.
- Connected PostgreSQL database for persistent storage of user and room data.

Authentication and User Management

- Implemented user signup and login functionality using JWT (JSON Web Token) for secure authentication.
- Created server-side validation and session handling for better user security.
- Linked the frontend authentication flow with backend routes for smooth login and room access.

Socket.io Integration

- Successfully established real-time socket connections between frontend and backend.
- Implemented event listeners for user join, leave, and room synchronization.
- Built the base structure for future real-time chat and status updates.

Room Management

- Developed a room creation and joining system where users can enter shared 3D spaces.
- Managed user sessions within rooms using socket events and backend validation.
- Added unique room codes and simple logic for room-based interaction segregation.

3D Environment Setup

- Designed and rendered a basic 3D meeting space with camera controls, lighting, and boundaries.
- Integrated user avatars into the environment for movement and visual presence.
- Applied perspective and collision controls to ensure smooth navigation.

Avatar Movement and Synchronization

- Enabled real-time avatar movement synchronization using socket events.
- Developed logic to broadcast user position and rotation updates to all connected clients.
- Created a simple but efficient movement control system compatible with multiple users.

UI/UX Improvements

- Added a clean interface for rooms, chat placeholders, and avatar controls.
- Enhanced responsiveness for different screen sizes using Tailwind utilities.
- Built a structured layout for future integration of chat, media controls, and user panels.

6. Work in Progress

The current phase of development in Real-Meet (3D Metaverse Communication Platform) is primarily focused on strengthening real-time communication, improving user interactivity, and enhancing the visual design of the 3D environment. Significant progress has been made in integrating sockets for user synchronization and interactions, while ongoing work continues to refine the system's responsiveness and realism.

Real-Time Communication (Socket.io)

- Implementing robust Socket.io events for real-time data transfer between clients and server.
- Features like chat messaging, hand-raising, and presence detection (user joining or leaving rooms) are under development.
- Ongoing optimization of socket connections to handle multiple users simultaneously with minimal latency.

Avatar Synchronization and Behaviour

- Working on smoother avatar movement synchronization across all connected clients.
- Adding real-time rotation and facing direction updates so avatars look toward the correct objects or users.
- Improving avatar responsiveness for a more realistic feel during interactions.

3D Environment Enhancement

- Refining the current 3D environment structure to make it visually appealing and interactive.
- Experimenting with camera controls, lighting adjustments, and collision handling for avatars.
- Designing a dynamic layout that supports multiple rooms or spaces.

User Interface Integration

- Developing and linking the chat interface to backend socket events.

- Enhancing the in-room UI elements, including user lists, status indicators, and message panels.
- Optimizing interface responsiveness for smoother transitions within the 3D view.

Backend and Database Connectivity

- Strengthening the Express.js backend for better request handling and socket management.
- Maintaining stable database operations for user sessions, room data, and chat history using PostgreSQL.
- Preparing for potential scaling by modularizing routes and controllers.

WebRTC Setup

- Setting Up the Web RTC in the frontend and signalling route in the backend to send the connectivity signals using secet.io.

7. Technology Stack

Programming Language:

- Javascript – main programming language for Both Frontend and Backend Development.

Frontend Frameworks:

- react.js – for Building of the frontend visible to users.
- Tailwind CSS –To design the frontend effectively.
- React-three-fibre & react-three-fibre/drei – To create 3D models and 3D Virtual Space.
- Framer-motion – for smooth animation

Backend libraries:

- Express– for creating a REST API to create API end Points.

Database:

- Postgres – for storing of user Information and credentials.

Communications:

- Express Routes - for GET, POST, DELETE, PUT requests from the frontend to the backend;
- Socket.io - For real-time two tier connection to handle real-time communication in the room.

Security:

- JWT – Jsonwebtoken for the authentication of the user and hold their sessions.
- Bcrypt.js – To encrypt password before storing to database for further security.

Hardware Requirements:

- Laptop/PC with 8 GB RAM
- Webcam for capturing Video

8. System Analysis and Design

Functional Requirements

- **User Authentication:** Secure login and signup system.
- **Room Creation:** Users can create, join, or leave 3D virtual spaces.
- **3D Avatars:** Each user is represented by a customizable 3D avatar.
- **Real-Time Communication:**
 - Live text chat using sockets.
 - Events like *hand raise*, *status updates*, or *user join/leave*.
- **Movement & Synchronization:**
 - Avatar movement and orientation updated in real time for all connected users.
- **Camera Control:**
 - Users can look around the 3D environment using orbit or first-person controls.

Non-Functional Requirements

- **Performance:** Maintain <200 ms latency for user movements and chat.
- **Scalability:** Support multiple rooms and concurrent users.
- **Security:** JWT-based authentication, secure cookies, and sanitized socket input.
- **Usability:** Responsive UI and intuitive 3D navigation.
- **Compatibility:** Works on modern browsers supporting WebGL.

System Architecture

Frontend (Client)

- Built using React.js and Three.js / React Three Fiber (R3F) for 3D rendering.
- Uses Socket.io Client to communicate with the backend in real time.
- Handles:
 - Rendering of 3D avatars and scenes.
 - Camera control and user input (keyboard, mouse).
 - Sending events for avatar movement and chat messages.

- Updating positions and actions of other users received via sockets.

Backend (Server)

- Developed using Node.js and Express.js.
- Uses Socket.io Server for handling live communication events such as:
 - joinRoom, leaveRoom, chatMessage, raiseHand, etc.
- Manages user sessions, room states, and broadcasting updates to all connected users.
- Provides REST APIs for user authentication and room data management.

Database Layer

- PostgreSQL database for persistent storage of:
 - User details (name, email, avatar data).
 - Room metadata (code, owner, creation time).
 - Chat logs and event history.

Rendering Layer

- Uses Three.js or React Three Fiber to render 3D spaces.
- Models and textures are either preloaded or fetched dynamically.
- Lights, cameras, and 3D assets are optimized for browser rendering

9. Pert Chart

The development of Real-Meet, a 3D metaverse-based communication platform, was planned and executed through multiple structured phases as represented in the PERT (Program Evaluation and Review Technique) chart. Each phase consists of interdependent activities that collectively lead toward the final implementation.

Phase 1: Project Initialization

The project began with problem identification, where the need for a browser-based metaverse platform enabling real-time interaction was established. This was followed by requirement analysis, involving both functional and non-functional aspects such as user management, 3D rendering, and real-time synchronization. After this, technology research was conducted to select suitable tools — React.js and Three.js for the frontend, Node.js and Socket.io for backend communication, and PostgreSQL for database management.

Phase 2: System Design

Based on the requirements, a system architecture was designed outlining the interaction between frontend, backend, and database components. This included defining API structures, socket communication flow, and event handling mechanisms. Next, a database schema was designed with tables for user authentication, room management, and socket sessions. Parallely, UI/UX design was conceptualized for the 3D environment, user avatars, and control interface.

Phase 3: Environment Setup

Once the design phase was complete, both frontend and backend environments were initialized. The React environment was configured with Tailwind CSS and React-Three-Fiber, while the backend was prepared using Express.js and database connectivity. After these setups, Socket.io was integrated to establish real-time communication between the client and server, marking the foundation for interactive features.

Phase 4: Core Development

The development phase began with the implementation of user authentication using JWT tokens to manage secure login and signup. Then, room management functionality was added to allow users to create, join, and leave virtual spaces. Subsequently, the 3D environment was created using Three.js, including the setup of cameras, lighting, and avatar placement. The next step was avatar movement synchronization, where socket events ensured that every user's movement was reflected in real time across all connected clients.

Currently, the real-time chat system is being developed using Socket.io for message broadcasting within rooms. Alongside, features such as hand-raising and basic interaction events (e.g., gestures or signals) are being integrated to enhance user interactivity within the metaverse.

Phase 5: Upcoming Tasks

In the upcoming phase, the focus will be on enhancing communication by integrating WebRTC for voice and video calling, improving 3D avatars and environments, and optimizing socket performance for scalability. Once these are completed, the system will undergo testing and deployment to platforms like Vercel (frontend) and Railway (backend).

10. Screenshots

1. Login Page

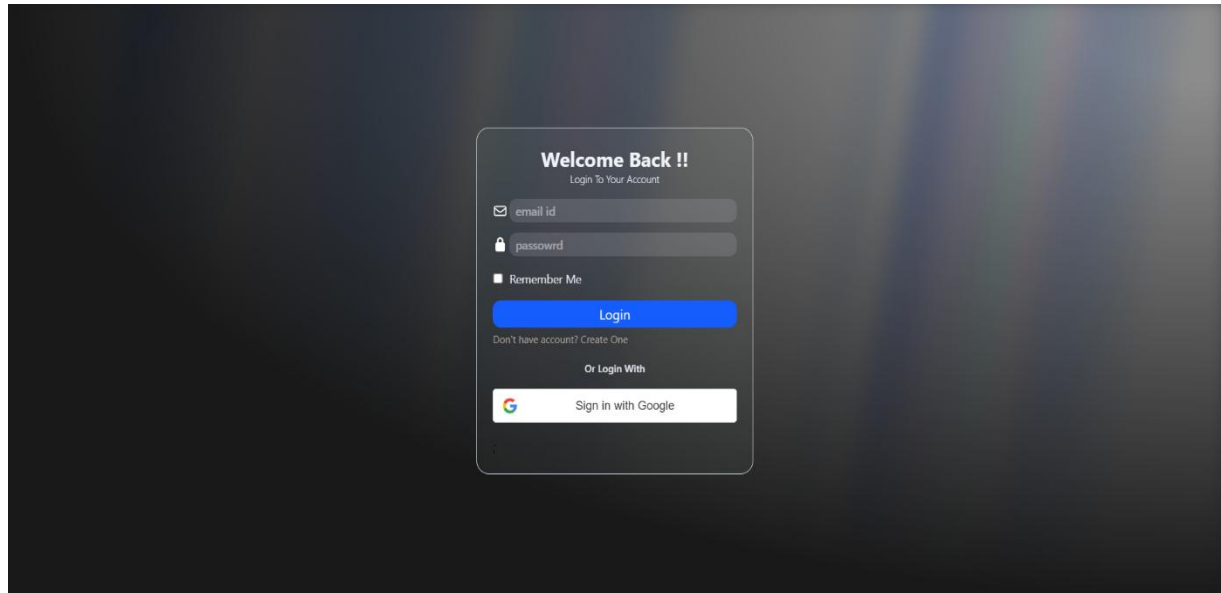


Fig.10.1 – Screenshot of login page

2. Signup Page

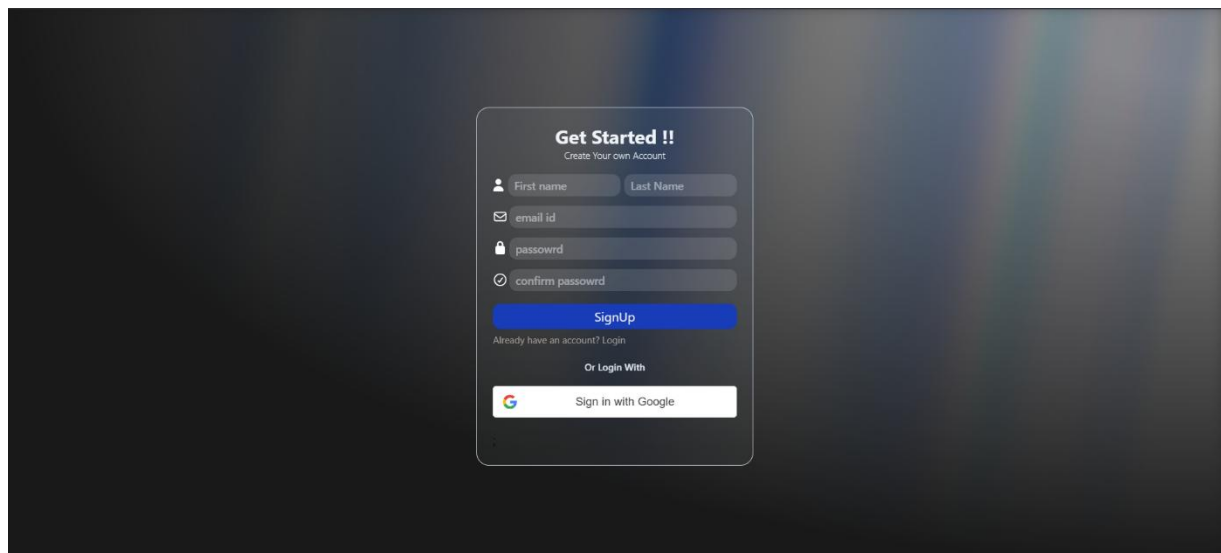


Fig.10.2 – Screenshot of signup page

3. Home Page

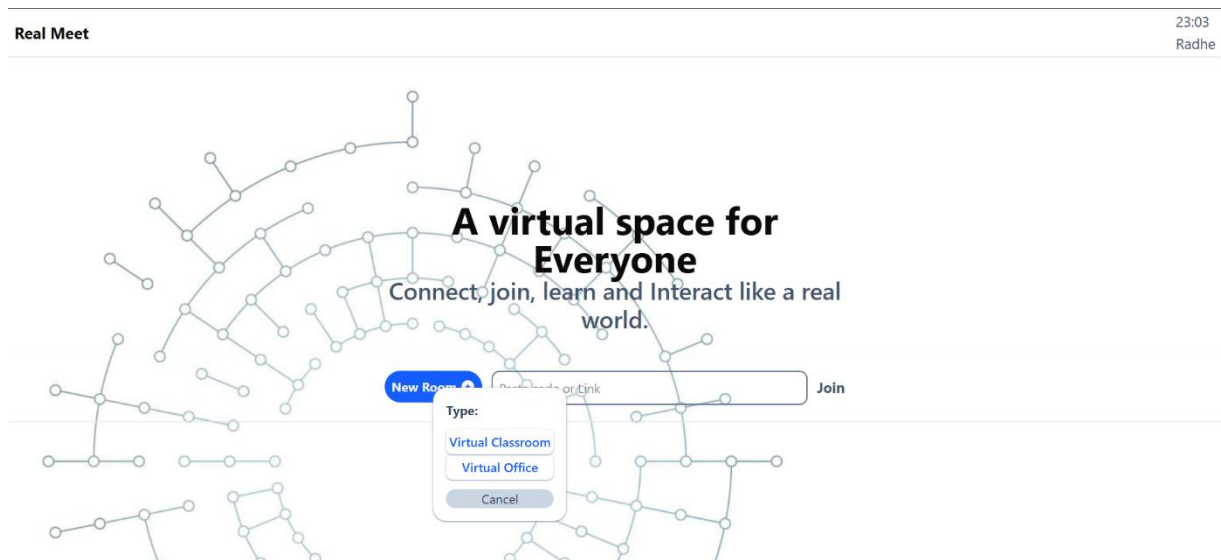


Fig.10.3 – Screenshot of Home page while creating Room

4. Virtual Meeting Room

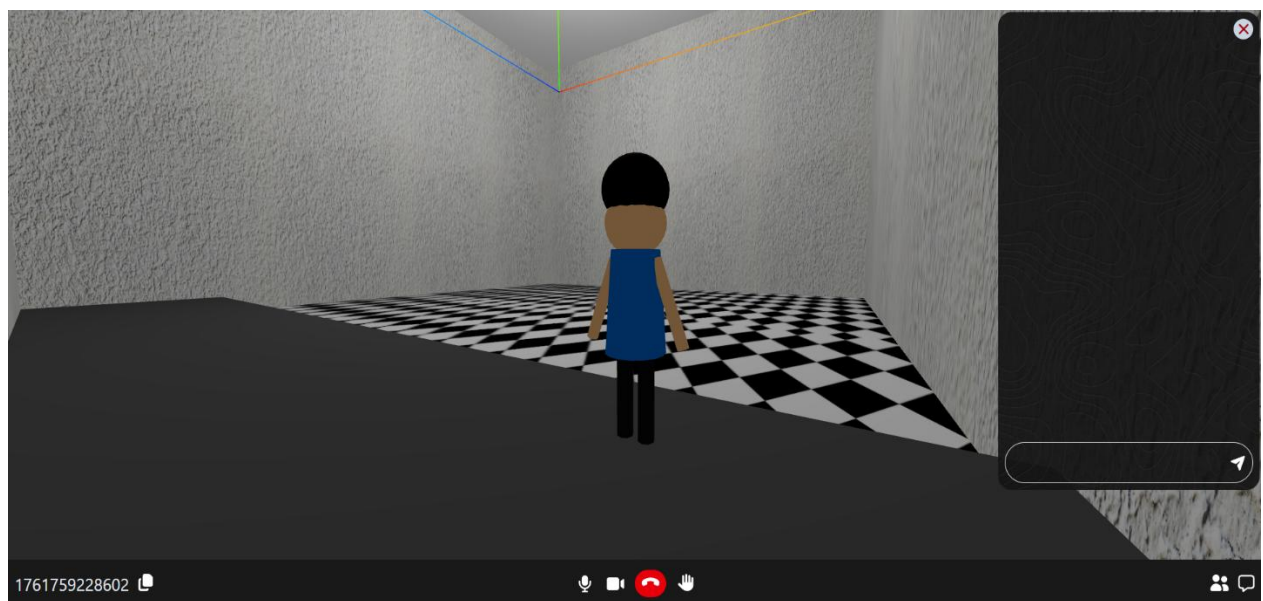


Fig.10.4 – Screenshot of 3D meeting room

11. Future Works

- **Avatar Realism and Interaction**

Enhance avatar design with improved 3D models, facial expressions, and movement animations to make interactions more natural and lifelike.

- **WebRTC Integration for Real-Time Communication**

Implement voice and video communication using WebRTC integrated with Socket.io for seamless signaling and session management.

- **Performance, Security, and Scalability**

Optimize socket event handling, ensure secure token-based authentication, and plan for scalability using tools like Redis or Socket.io clustering.

- **Testing, Deployment, and Future Enhancements**

Conduct extensive testing, deploy updates to Vercel and Railway, and explore advanced features like spatial audio, private rooms, and AI-powered avatars.

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